



HINDUSTAN COPPER LIMITED
1 Ashutosh Chowdhury Avenue
Kolkata - 700019
CIN: L27201WB1967GOI028825

Invites

EXPRESSION OF INTEREST (EOI)
FROM
TECHNOLOGY PROVIDERS/LICENSORS

FOR

**MANUFACTURING OF COPPER CATHODE BY PROCESSING SULPHIDE
COPPER CONCENTRATE THROUGH HYDRO-METALLURGY ROUTE**

Ref No. : HCL/HM/EOI/2016

Due Date : 04.04.2016, 15 :00 Hrs.

PRESS RELEASE

HINDUSTAN COPPER LIMITED

Notice Inviting Expression of Interest (Eoi)

Reference No: HCL/HM/EOI/2016

Date: 26.2.2016

Hindustan Copper Limited (HCL) invites Expression of Interest from potential Technology Providers / Licensors for manufacturing of copper cathode by processing sulphide copper concentrate (chalcopyrite) through hydrometallurgy route. Subsequent addendum/corrigendum of the Eoi shall be posted/updated in the HCL website.

For details of complete document of EOI, Interested parties may refer HCL's Website www.hindustancopper.com.

Due Date & Time for submission of EOI: 4.04.2016, 15:00 hrs. (Indian Standard Time).

Email: santoshsharma@hindustancopper.com

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Invitation for Global Expression of Interest

1.0 INTRODUCTION

Hindustan Copper Limited (HCL), a Mini Ratna Public Sector Enterprise under the administrative control of the Ministry of Mines, was incorporated on 9th November 1967. It has the distinction of being the nation's only vertically integrated copper producing company as it manufactures copper right from the stage of mining to beneficiation, smelting, refining and casting of refined copper metal into downstream saleable products. HCL has the distinction of owning all copper mining leases in the country.

The company has its Corporate and Registered office in Kolkata. HCL has five Units viz. Khetri Copper Complex (Rajasthan), Malanjkhand Copper Project (Madhya Pradesh), Indian Copper Complex, Ghatsila (Jharkhand), Taloja Copper Project (Maharashtra) and Gujarat Copper project (GCP) Gujarat. The marketing/sales setup is headquartered at Kolkata with four regional sales offices in Delhi, Mumbai, Kolkata & Bangalore and sales godowns spread across the country.

The company manufactures and markets copper cathodes, copper continuous cast copper wire rod and by-products such as anode slime (containing gold, silver & selenium), copper sulphate, slime and sulphuric acid.

Further details about the company and its operations can be obtained from its website www.hindustancopper.com.

2.0 INTENT OF EOI:

HCL is implementing an expansion programme to enhance to quadruple its mine capacity from 3.2 to 12.4 million tonne per annum. Accordingly, copper in concentrate production of the Company will increase from 32,000 tonne to around 125,000 tonne in next four years. The bulk of the concentrate production will come from Malanjkhand Copper project where it is implementing an underground copper mine of capacity 5 million tonne per annum and balance from mines of Khetri copper complex at Rajasthan. The mineralogy of copper concentrate of Malanjkhand & Khetri is given at annexure-1.

The Company plans to process the above copper concentrate to refined metal through a cost effective atmospheric pressure level leaching, solvent extraction and

subsequent electro-winning route i.e. through a hydro-metallurgical extraction technology.

3.0 Invitation of the Proposal:

Hindustan Copper Ltd invites response from suppliers of technology for conversion of chalcopryrite concentrate to LME grade “A” cathode copper through hydro-metallurgical route which shall comprise atmospheric pressure level leaching, for a plant of 1 lakh tonne capacity. The technology should be capable of recovering high grade sterling silver and gold of bullion grade from the concentrate. The package should include technology for conversion of all process residues to stable and non-hazardous material. The technology supplier should be able to provide extensive process guarantees & detailed basic engineering for the entire process including flow sheets, list of equipments along with their detailed specifications. The prospective bidders need to indicate the CAPEX and OPEX at +/- 30% accuracy for the proposed facility on the basis of detailed analysis of input material as mentioned below:

The bidders are to include the requirement of land for auxiliary equipments and for storage of environmentally stable waste residue.

4.0 Submission of Proposal: The proposal(s) is to be submitted giving following details of the item below:

S.No	Item
I	General Details of the Bidder:
	Name of the BIDDER / Firm submitting EOI
	Nature or legal status of the Firm
	Name and address of associated companies to be involved in the project with relationship and role, if any.
	Registered Address of Firm
	Designation and address of Contact person
	Turnover & Net worth of the Firm during last three financial years (Please enclose copy of audited annual reports)
	In case firm is a subsidiary, indicate the role of parent company
II	Technical Information of Technology Provider/Licensors
	Details of plants - Provided technology/ services/license Demo & Commercial both: i. Ownership ii. Location iii. Operational since iv. Plant capacity v. Plant performance
III	a. Proposed Technology and Business Model for setting up leaching , solvent extraction & elctro-winning plant to produce 100,000 MT/year of copper cathode (LME- Grade “A”) in India.

	b. Mention details of the recovery to be obtained from the throughput of the final product & by products. c. Indicative Capital cost of the Plants with break-up under major heads of expenditure. d. Indicative Production cost per tonne of cathode produced e. Scheme for the disposal of the effluent generated. f. Comparison of proposed technology with other available technologies like pyro smelting and electro refining of anode, sulphate based chalcopryrite leaching, SX-EW process, chloride based Chalcopryrite leaching SX-EW process, Ammonia based chalcopryrite leaching, SX-EW process, etc.
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5.0 Pre-Qualification Criteria:

5.1 Technology/Technical Criteria:

- i. Technology should be proven in minimum in continuous pilot /commercial scale. The technology should be flexible to scale up. The proposed process technology should be flexible to utilize variable quality sulphide (mainly chalcopryrite) copper concentrate. The technology provider should have either have obtained patent or applied for the technology proposed in the EoI.
- ii. The Technology Provider should be capable of providing Basic Design Package of the proposed technology to convert sulphide (mainly chalcopryrite) copper concentrate to copper cathode through hydro-metallurgy route. Provide list of proposed deliverables of basic design package.
- iii. The proposed conversion technology provider shall have at least one proven operating plant or pilot scale testing with processing the sulphide (mainly chalcopryrite) copper concentrate successfully on continuous basis in India/abroad.
- iv. The technology provider should also be prepared to provide guarantee of the plant in terms of its product quality and quantity, metal recovery, plant reliability, OPEX, etc. An undertaking to that extent is required to be submitted for becoming eligible to be shortlisted.

5.2 Financial Criteria

Net Worth: Net Worth of the Company should be positive as per the immediate preceding year's audited financial results.

6.0 General:

- i. Language of EOI and Communication shall be in English. In case any document is submitted in other than English language, authenticated English translation of the same shall be enclosed.
- ii. The BIDDER must not have been debarred/ blacklisted by any Central/State Govt. department, agency, PSUs/ Institution/ Agencies/ Autonomous organization of India. The BIDDER shall submit a self-certification by an authorized person to this effect.
- iii. All revisions, clarifications, corrigenda, addenda, time extensions etc., to the EOI will be hosted only on HCL website www.hindustancopper.com. Bidders should regularly visit these websites to keep themselves updated.
- iv. HCL will not be responsible for any cost or expenses incurred by the BIDDER in connection with preparation or delivery of EOI.
- v. HCL reserves the right to call the BIDDER for presentation at its office located in Kolkata, India and may visit the sites of the BIDDER. The BIDDER shall have no objection whatsoever in this regard. HCL will not be responsible for any cost or expenses incurred by the BIDDER for visiting its office at for meeting & presentation.
- vi. HCL will be at liberty to amend/modify/delete any of the above conditions and/ or the process as may be deemed necessary in the light of the facts and circumstances of the case.

7.0 Eoi Process

Following completion of the initial process, all short listed parties, before submission of bids will be provided with an opportunity to hold further discussions with the senior management of HCL. They will also be furnished with further information, if required. However, information will be provided at the sole discretion of HCL. All short listed parties before submission of bids will be provided with an opportunity to:

- (i) Conduct a technical and financial due diligence;
- (ii) Hold further discussions with the senior management of HCL; and
- (iii) Undertake mine/plant visits.

The terms relating to submission of bids will be set out in the RFP and other document(s) to be prepared for the purpose.

8.0 IMPORTANT NOTE:

This advertisement does not constitute and will not be deemed to constitute any commitment on part of HCL to set up the copper cathode manufacturing plant through hydro-metallurgy route in India. HCL has the sole discretion to qualify or accept the EOI and reject the proposal without assigning any reason whatsoever. Furthermore, this advertisement neither confirms the right nor an expectation on any party to participate in the proposed EOI.

Clarification, if any, can be obtained from Mr. Santosh Sharma, Executive Director (Operations)/Mr. R.Adhikari, Chief Manager (Chemical) & Mr. Vivek Gupta, Chief Manager (IE) through following E-mail:

- i. santoshsharma@hindustancopper.com/+919433001146
- ii. vivek_g@hindustancopper.com/Mob:+919051136761;
ramananda_a@hindustancopper.com/mob : +91 9051306932
- iii. hindcopper@hotmail.com

9.0 Date & Time for Submission of Proposal (EOI):

The prospective bidders are required to send the EOI document along with all supporting documents and duly signed & stamped on each page (in original +one copy) so as to reach at the following address within the due date & time. The due date & time for submission of Proposal (EOI) is 4.4.2016,1500 hrs. (Indian Standard Time).

Hindustan Copper Limited
Operations Department
1, Ashutosh Chowdhury Avenue
Kolkata - 700019
Attn: Shri Santosh Sharma, Executive Director (Operations)

Form 1

Submission of Bid

(To be forwarded on the letter head of the interested party submitting the Eoi)

Ref :

Date

To
Shri Santosh Sharma
Executive Director(Operations)
Hindustan Copper Limited
1 Ashutosh Chowdhury Avenue
Kolkata - 700 019
INDIA

Sub: Manufacturing of copper cathode by processing sulphide copper concentrate through hydro-metallurgy route.

Please refer to the invitation for Expression of Interest issued in respect of the above mentioned subject matter. We have read and understood the contents of the invitation for EOI. We are desirous of selection as preferred technology supplier for manufacturing of copper cathode by processing sulphide copper concentrate through hydro-metallurgy route and propose to submit this EOI.

We believe that our Company satisfies the eligibility criteria set out in the relevant sections of the EOI. The information template and litigation impact statement of legal capacity as per the format indicated in the EOI duly signed by us are enclosed.

You and your authorized representatives may contact the person(s) listed below for further information. The undersigned is (are) fully authorized to act on behalf of the respondent

We shall be glad to receive further communication on the subject.

Signature

Name

(For <Respondent>)

Form 2

Litigation Impact Statement

(To be forwarded on the letter head of the interested party submitting the Eol)

Ref :

Date

To
Shri Santosh Sharma
Executive Director(Operations)
Hindustan Copper Limited
1 Ashutosh Chowdhury Avenue
Kolkata - 700 019
INDIA

Sub: Manufacturing of copper cathode by processing sulphide copper concentrate through hydro-metallurgy route.

Sirs,

This is in reference to the Expression of Interest of HCL for Manufacturing of copper cathode by processing sulphide copper concentrate through hydro-metallurgy route. We have read and understood the contents of the Eol document and pursuant to it, hereby confirm that we satisfy the eligibility criteria laid out in the Eol.

We hereby confirm that save as may be set out in the schedule attached to this statement, there is no litigation (including court, arbitration and other proceedings), inquiry or order from any regulatory authority, current or pending against us, which if adversely determined might have material adverse impact on our ability to carry on business or pay our debts as they fall due or on our ability to enter into any of the transactions contained in, or contemplated, in respect of the process of identification of an Interested Party for Manufacturing of copper cathode by processing sulphide copper concentrate through hydro-metallurgy route. Further, we have not been debarred/ blacklisted by any Central/State Govt. department, agency, PSUs/ Institution/ Agencies/ Autonomous organization of India

The undersigned is authorized signatory and is vested with requisite powers to furnish such a letter and Request for Qualification and authenticate the same

Yours faithfully,
(_____)
Authorised signatory
for and on behalf of

Form 3**DETAILS OF NET WORTH**

FINANCIAL YEAR	DETAILS OF NET WORTH (INR)			
	PAID UP CAPITAL (a)	FREE RESERVE (b)	ACCUMULATED LOSS (c)	NET WORTH (a+b-c)
As per immediate preceding financial year				

Annual Report / Financial statement for the immediate preceding financial year shall be submitted along with EOI.

Appendix - Definitions

In this Eol document, the following terms shall be defined as indicated:

- i. “HCL” means Hindustan Copper Limited;
- ii. “Eol” means an expression of interest made in response to this ;
- iii. “Bidder” means The Technology provider/Licensors.
- iv. Respondent” means any entity, including a company or firm which returned a Cover Letter to the Contact Person, and plans to submit an Eol in response to this Eol.
- v. “Copper Cathode” means electrolytic grade conforming to specification of LME”A” grade.

Modal Mineralogy of Khetri and Malanjkhand Ore Samples

Mineral	Khetri feed - Wt%	Malanjkhand feed - Wt%
Chalcopyrite	2.70	3.86
Cubanite	0.08	0.00
Chalcocite	0.00	0.07
Digenite	0.00	0.03
Covellite	0.00	0.02
Sphalerite	0.00	0.01
Pyrite	0.45	1.36
Pyrrhotite	6.08	0.00
Molybdenite	0.00	0.01
Quartz	39.61	81.89
Plagioclase	4.00	2.54
K-feldspar	0.14	4.37
Epidote	0.00	1.02
Tourmaline	0.90	0.15
Almandine	8.64	0.04
Ferrosilite	3.52	0.01
Tremolite	0.25	0.00
Actinolite	0.28	0.24
Ferroactinolite	0.13	0.00
Cummingtonite	4.42	0.01
Anthophyllite	0.96	0.00
Hornblende	1.65	0.14

Ferropargasite	1.65	0.04
Pyrochlore	0.01	0.02
Biotite	6.29	0.21
Chlorite	12.30	1.91
Muscovite	0.40	0.11
Kaolinite	0.04	0.10
Clay	0.13	0.02
Titanite	0.00	0.03
Ferropyrosmalite	0.13	0.00
Marialite	0.37	0.02
Andalusite-Kyanite	0.21	0.00
Allanite	0.11	0.03
Calcite	0.11	0.05
Ankerite	0.70	0.00
Corundum	0.00	0.00
Apatite	0.08	0.05
Magnetite	2.45	1.03
Ilmenite	0.38	0.01
Goethite	0.31	0.17
Native copper	0.00	0.03
Iron	0.06	0.05
Unknown	0.45	0.35
Total	100.00	100.00